

1. (35 points) Find the general solution of

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Soln Taking Laplace transform at both sides;

$$\mathcal{L}\{y''\} + 5\mathcal{L}\{y'\} + 6\mathcal{L}\{y\} = \mathcal{L}\{e^{-2x}\}$$

$$\text{Let } \mathcal{L}\{y\} = Y(s)$$

$$s^2 Y(s) - sy(0) - y'(0) + 5sY(s) - 5y(0) + 6Y(s) = \frac{1}{s+2}$$

$$\Rightarrow Y(s)(s^2 + 5s + 6) = \frac{1}{s+2} + (s+5)y(0) + y'(0)$$

$$\Rightarrow Y(s) = \frac{1}{(s+2)^2(s+3)} + \frac{(s+5)y(0) + y'(0)}{(s+2)(s+3)}$$

$$\text{Let } \frac{1}{(s+2)^2(s+3)} = \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{s+3} \quad \Rightarrow \begin{cases} A+C=0 \\ 5A+B+C=0 \\ 6A+B+C=1 \end{cases}$$

$$\text{Let } \frac{(s+5)y(0) + y'(0)}{(s+2)(s+3)} = \frac{C_1}{s+2} + \frac{C_2}{s+3}$$

Since C_1 and C_2 will depend on $y(0)$ and $y'(0)$ we'll keep them as arbitrary constants

$$\text{Then } \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{-1}{s+2} + \frac{1}{(s+2)^2} + \frac{1}{s+3} + \frac{C_1}{s+2} + \frac{C_2}{s+3}\right\}$$

$$\Rightarrow y(x) = -e^{-2x} + xe^{-2x} + e^{-3x} + C_1 e^{-2x} + C_2 e^{-3x}$$

$$\Rightarrow y(x) = D_1 e^{-2x} + D_2 e^{-3x} + x e^{-2x}$$

2. (35 points) Find the lowest order homogeneous linear differential equation with real and constant coefficients which admits $e^x \cos 2x$ and x^2 as some of its solutions.

Since $e^x \cos 2x$ is one of solutions, the characteristic equation must have the complex numbers $(1+2i)$ & $(1-2i)$ as roots. So we have $(r-(1+2i))(r-(1-2i))$ term.

Another solution is x^2 . So the characteristic equation must have the r^3 term.

So, the lowest order:

$$p(r) = r^3(r - (1+2i))(r - (1-2i)) = r^3(r^2 - 2r + 5) \\ = r^5 - 2r^4 + 5r^3$$

$$y^{(5)} - 2y^{(4)} + 5y^{(3)} = 0$$

$$\frac{d^5 y}{dt^5} - 2 \frac{d^4 y}{dt^4} + 5 \frac{d^3 y}{dt^3} = 0$$

3. (30 points) Transform the equation

$$tu'' + t^2u' + (3t^2 - t)u = e^t$$

into a linear system of first order differential equations.

let $x_1 = u, x_2 = u'$

$$x_1' = u' = x_2$$

$$x_1' = x_2$$

$$tx_2' + t^2x_2 + (3t^2 - t)x_1 = e^t \rightarrow x_2' + tx_2 + (3t - 1)x_1 = e^t/t$$

$$\begin{cases} x_1' = x_2 \\ x_2' = e^t/t - tx_2 - (3t - 1)x_1 \end{cases}$$

matrix form

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -(3t-1) & -t \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ e^t/t \end{bmatrix}$$

$\vec{x}' = \vec{A} \vec{x} + \vec{f}(t)$